INCH-POUND

MIL-P-70543B (AR)
16 August 1994

SUPERSEDING
MIL-P-70543A (AR)
22 January 1988

MILITARY SPECIFICATION

PROJECTILE, 81MM, ILLUMINATING, M853A1
PROJECTILE, 81MM, INFRARED ILLUMINATING, XM816
METAL PARTS

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, examinations and tests for metal parts and sub-assemblies for the projectile, 81mm, Illuminating, M853A1 and Infrared, Illuminating, XM816.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315
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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9381093 - Projectile, 81MM: Illuminating, M853A1 Loading Assembly
12953388 - Projectile, 81MM: Infrared Illuminating, XM816 Loading Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications, and this specification.

3.2 Components and assemblies. The components and assemblies shall comply with the applicable requirements specified on Drawing (dwg.) 9381093 and 12953388 and associated drawings and with all requirements specified in applicable specifications, standards, and this specification.

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Wood insert. The moisture content of the wood insert shall not exceed six percent for use on the XM816 Infrared Illuminating Cartridge.

3.5 Workmanship. Threads shall be full and undamaged for the entire minimum length or depth. Threads and other parts shall be free of burrs or foreign matter which may interfere with further assembly or proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated with residue. Wood insert for the XM816 Canister loading shall be free of cracks, holes, cavities or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see TABLE I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile, 81MM: Illuminating M853A1 Assembly and Proj, 81MM: Infrared Illum XM816 Assembly	SHEET 1 OF 2		DRAWING NUMBER See below
				NEXT HIGHER ASSEMBLY N/A
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cage, Bearing (Dwg. 9381087) Examination for defects	15	3.2	4.4.2.1
	Canister, Illuminant (Dwg. 9381076) Examination for defects	15	3.2	4.4.2.2
	Plate, Base (Dwg. 9381086) Examination for defects	15	3.2	4.4.2.3
	Pin (Dwg. 9381088) Examination for defects	15	3.2	4.4.2.4
	Sleeve, Split (Dwg. 9369959) Examination for defects	15	3.2	4.4.2.7
	Plate, Droque (Dwg. 9369956) Examination for defects	15	3.2	4.4.2.8
	Plate, Support (Dwg. 9381085) Examination for defects	15	3.2	4.4.2.10
	Tube, Body (Dwg. 9381090) Examination for defects	15	3.2	4.4.2.11
	Adapter (Dwg. 9381091) Examination for defects	15	3.2	4.4.2.12

NOTES:

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile, 81MM: Illuminating M853A1 Assembly and Proj, 81MM: Infrared Illum XM816 Assembly	2 OF 2	See below
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Body Tube Assembly (Dwg. 9349905) Examination for defects	15	3.2	4.4.2.13
	Cone, Tail (Dwg. 9327836) Examination for defects	15	3.2	4.4.2.14
	Parachute, Flat Circular (Dwg. 9381631) Examination for defects	15	3.2	4.4.2.15
	Parachute Folding and Packing Assembly (Dwg. 9381075) Examination for defects	15	3.2	4.4.2.16
	Bag, Deployment (Dwg. 9381589) Examination for defects	15	3.2	4.4.2.17
	Swivel Assembly (Dwg. 9381083) Examination for defects	15	3.2	4.4.2.18
	Ignition Disc Assembly (Dwg. 9381078) Examination for defects	15	3.2	4.4.2.19
	Canister Illuminant Assembly (Dwg. 9381077) Examination for defects	15	3.2	4.4.2.20
	Wire, Suspension (Dwg. 9381073) Examination for defects	15	3.2	4.4.2.21

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.4). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70543B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cage, Bearing	1 OF 1	9381087
			NEXT HIGHER ASSEMBLY
			9381083

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness of plate	Level III	3.2	Gage
102	Diameter of cage	Level III	3.2	Gage
103	Diameter of well	Level III	3.2	Gage
104	Diameter of thru-hole	Level III	3.2	Gage
105	Depth of well	Level III	3.5	Gage
Minor				
201	Radius, incorrect	Level V	3.2	Gage
202	Surface finish improper	Level V	3.2	Visual/4.5.4
203	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70543B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Canister, Illuminant	1 OF 1	9381076
			NEXT HIGHER ASSEMBLY
			9381077

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wall thickness, min.	Level III	3.2	Gage
102	Diameter of thru-hole, max.	Level III	3.2	Gage
103	Inside diameter of canister, min.	Level III	3.2	Gage
104	Length of canister	Level III	3.2	Gage
105	Hardness	Level III	3.5	4.5.1
Minor				
201	Protective finish missing or inadequate	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70543B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Plate, Base	1 OF 1	9381086
			NEXT HIGHER ASSEMBLY: 9381083

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of plate	Level III	3.2	Gage
102	Thickness of plate	Level III	3.2	Gage
Minor				
201	Surface finish improper	Level V	3.2	Visual/4.5.4
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Pin	1 OF 1	9381088
			NEXT HIGHER ASSEMBLY: 9381083

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter above base	Level III	3.2	Gage
102	Height of base	Level III	3.2	Gage
Minor				
201	Outside diameter of base of pin	Level V	3.2	Gage
202	Location of hole on center of pin	Level V	3.2	Gage
203	Distance from top to center of hole, min.	Level V	3.2	Gage
204	Diameter of thru-hole, min.	Level V	3.2	Gage
205	Overall height	Level V	3.2	Gage
206	Surface finish improper	Level V	3.2	Visual/4.5.4
207	Inside radius, max.	Level V	3.2	Gage
208	Evidence of poor workmanship	Level V	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.5	Ring, Filler	SHEET 1 OF 1	9381080
			NEXT HIGHER ASSEMBLY: 9381078

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Hole missing	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Spacer, Felt	SHEET 1 OF 1	9381074
			NEXT HIGHER ASSEMBLY: 9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Hole missing	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Sleeve, Split	1 OF 1	9369959
			NEXT HIGHER ASSEMBLY
			9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length of sleeve	Level III	3.2	Gage
102	Thickness of sleeve, max.	Level III	3.2	Gage
Minor				
201	Outside radius of sleeve	Level V	3.2	Gage
202	Arc length	Level V	3.2	Gage
203	Perpendicularity of ends	Level V	3.2	Gage
204	Surface finish improper	Level V	3.2	Visual/4.5.4
205	Evidence of poor workmanship	Level V	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Plate, Drogue	1 OF 1	9369956
			NEXT HIGHER ASSEMBLY
			9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of plate, max.	Level III	3.2	Gage
102	Thickness of plate, min.	Level III	3.2	Gage
Minor				
201	Diameter of thru-hole, min.	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Cup, Parachute	1 OF 1	9369955
			NEXT HIGHER ASSEMBLY: 9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside radius of cutout	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Plate, Support	1 OF 1	9381085
			NEXT HIGHER ASSEMBLY: 9381077

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, max.	Level III	3.2	Gage
102	Inside diameter, min.	Level III	3.2	Gage
103	Thickness	Level III	3.2	Gage
104	Flatness	Level III	3.2	Gage
105	Hardness	Level III	3.2	Gage/4.5.2
Minor				
201	Surface finish	Level V	3.2	Visual/4.5.4
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Tube, Body	1 OF 2	9381090
			NEXT HIGHER ASSEMBLY: 9349905

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material damaged or defective	Level III	3.1	Visual
102	Forward outside diameter	Level III	3.2	Gage
103	Rear outside diameter	Level III	3.2	Gage
104	Rear inside diameter	Level III	3.2	Gage
105	Forward inside diameter	Level III	3.2	Gage
106	Central inside diameter	Level III	3.2	Gage
107	Depth of tail cone well	Level III	3.2	Gage
108	Depth of fuze adapter well	Level III	3.2	Gage
109	Distance from front face to start of rear O.D., max.	Level III	3.2	Gage
110	Overall length	Level III	3.2	Gage
111	Runout of rear inside diameter with front face	Level III	3.2	Gage
112	Runout of rear inside diameter to rear face	Level III	3.2	Gage
113	Runout of forward inside diameter with rear inside diameter and rear face	Level III	3.2	Gage
114	Runout of forward outside diameter with rear inside diameter and rear face	Level III	3.2	Gage
115	Runout of rear inside diameter with rear outside diameter	Level III	3.2	Gage

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Tube, Body (Cont'd.)	2 OF 2	9381090
			NEXT HIGHER ASSEMBLY: 9349905

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Surface finish improper	Level V	3.2	Visual/4.5.4
202	Angle rear of forward bourrelet	Level V	3.2	Gage
203	Incorrect chamfer at front face	Level V	3.2	Gage
204	Presence of radius at rear end	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Adapter	1 OF 1	9381091
			NEXT HIGHER ASSEMBLY: 9349905

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Forward outside diameter	Level III	3.2	Gage
102	Rear outside diameter	Level III	3.2	Gage
103	Runout of front diameter with rear diameter and front shoulder	Level III	3.2	Gage
104	Perpendicularity of shoulder to rear outside diameter	Level III	3.2	Gage
105	Parallelism of front face with shoulder	Level III	3.2	Gage
106	Pitch diameter of threads	Level III	3.2	Gage
107	Minor diameter of threads, max.	Level III	3.2	Gage
108	Distance from front face to shoulder	Level III	3.2	Gage
109	Overall length	Level III	3.2	Gage
Minor				
201	Diameter at front of forward chamfer	Level V	3.2	Gage
202	Diameter at rear of rear chamfer	Level V	3.2	Gage
203	Radius or chamfer missing or incorrect	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Body Tube Assembly	1 OF 2	9349905
			NEXT HIGHER ASSEMBLY: 9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealant missing or inadequate on mating surfaces	Level III	3.2	Visual
102	Diameter of retaining pin holes	Level III	3.2	Gage
103	Depth of retaining pin holes	Level III	3.2	Gage
104	Retaining pin(s) missing	Level III	3.2	Visual
105	Retaining pin(s) not flush or below	Level III	3.2	Visual/Tactile
106	Length over maximum	Level III	3.2	Gage
107	Excessive gap between body tube and adapter	Level III	3.2	Gage
108	Runout of thread pitch diameter at projected length to rear inside diameter and rear and front faces	Level III	3.2	Gage
109	Diameter over max, after paint	Level III	3.2	Gage
110	Protective finish present on rear inside diameter (I.D.) or adapter threads	Level III	3.2	Visual
111	Corrosion preventative compound missing or not full on threads	Level III	3.2	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.13	Body Tube Assembly (Cont'd.)	SHEET 2 OF 2	9349905
			NEXT HIGHER ASSEMBLY
			9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Marking incorrect, missing or illegible	Level V	3.2	Visual
202	Protective finish missing or not full	Level V	3.2	Visual
203	Distance of retaining pin holes to face of adapter	Level V	3.2	Gage
204	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Cone, Tail	1 OF 2	9327836
			NEXT HIGHER ASSEMBLY
			9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of rear thread	Level III	3.2	Gage
102	Perpendicularity of rear face with pitch diameter (PD) of rear thread	Level III	3.2	Gage
103	Runout at rear of taper with forward OD	Level III	3.2	Gage
104	Runout at front of taper with forward OD	Level III	3.2	Gage
105	Runout of obturator groove with bourrelet	Level III	3.2	Gage
106	Runout of bourrelet with forward OD	Level III	3.2	Gage
107	Runout of forward inside diameter (ID) with forward OD	Level III	3.2	Gage
108	Wall thickness	Level III	3.2	Gage
109	Obturator groove diameter	Level III	3.2	Gage
110	Bourrelet diameter after paint, max.	Level III	3.2	Gage
111	Forward ID	Level III	3.2	Gage
112	Pitch diameter of rear thread	Level III	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Cone, Tail (Cont'd.)	2 OF 2	9327836
			NEXT HIGHER ASSEMBLY: 9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
113	Minor diameter of rear thread, max.	Level III	3.2	Gage
114	Distance from front face to shoulder	Level III	3.2	Gage
115	Forward OD	Level III	3.2	Gage
116	Overall length	Level III	3.2	Gage
117	Distance from front face to bourrelet	Level III	3.2	Gage
118	Width of obturator groove	Level III	3.2	Gage
119	Perpendicularity of forward shoulder with forward OD	Level III	3.2	Gage
Minor				
201	Distance from front face to forward taper diameter	Level V	3.2	Gage
202	Distance from front face to rear taper diameter	Level V	3.2	Gage
203	Protective finish improper	Level V	3.2	Visual
204	Radius incorrect or blends discontinuous	Level V	3.2	Visual
205	Chamfer incorrect	Level V	3.2	Gage
206	Surface finish improper	Level V	3.2	Visual/4.5.4
207	Distance from rear face to base of cavity, min.	Level V	3.2	Gage
208	Angle of obturator groove	Level V	3.2	Gage
209	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Parachute, Flat Circular	1 OF 1	9381631
			NEXT HIGHER ASSEMBLY: 9381075

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hole or tear in canopy greater than 1/8 inches	Level III	3.2	Gage
102	One or more shroud line(s) missing	Level III	3.2	Visual
103	Stitching missing or broken	Level III	3.2	Visual
104	Loop missing or improperly taped	Level III	3.2	Visual
105	Diameter of vent hole	Level III	3.2	Gage
106	Missing reinforcing tape	Level III	3.2	Visual
107	More than one patch	Level III	3.2	Visual
108	The patched hole or tear exceeds 1/2 inch max	Level III	3.2	Gage
109	Location bartack	Level III	3.2	Gage
110	Tape to hold five sets of shroud lines together missing	Level III	3.2	Visual
111	Length of stitched shroud lines together to form a loop	Level III	3.2	Gage
Minor				
201	Length of shroud line at attachment hem, min.	Level V	3.2	Gage
202	Length of reinforcing tape at shroud line attachment hem, min.	Level V	3.2	Gage
203	Developed diameter of canopy	Level V	3.2	Gage
204	Length of shroud lines from canopy hem to confluence point	Level V	3.2	Gage
205	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.16	Parachute Folding and Packing Assembly	SHEET 1 OF 1	9381075
			NEXT HIGHER ASSEMBLY
			9381093/12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Parachute bag stitching missing or broken	Level III	3.2	Visual
102	Parachute bag pull cord missing or improperly attached	Level III	3.2	Visual
103	Improper accordian fold	Level III	3.2	Visual
104	Shroud lines twisted or tangled or 1/2 of shroud lines not coiled under the last fold	Level III	3.2	Visual
Minor				
201	String missing or improperly tied	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Bag, Deployment	1 OF 1	9381589
			NEXT HIGHER ASSEMBLY: 9381075

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside diameter of bag, min.	Level V	3.2	Gage
202	Inside depth of bag, min.	Level V	3.2	Gage
203	Stitching incorrect or broken	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.18	Swivel Assembly	SHEET 1 OF 1	9381083
			NEXT HIGHER ASSEMBLY 9381084

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Swivel pin fails to rotate freely	Level III	3.2	Visual/Manual
102	Swivel assembly fails to pass through a 2.633 diameter ring gage	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Ignition Disc Assembly	1 OF 1	9381078
			NEXT HIGHER ASSEMBLY
			9381079/12953387

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Missing gauze	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Canister Illuminant Assembly	1 OF 1	9381077
			NEXT HIGHER ASSEMBLY: 9381084

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Canister illuminating assembly fails to pass through the ring gage	Level III	3.2	Gage
102	Spot weld improper or missing	Level III 1/	3.2	Visual/Manual, 4.5.3
Minor				
201	Length	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

1/ In addition to the visual inspection, five (5) parts from each lot shall be tested for adequate weld strength per para. 4.5.3.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Suspension Wire	1 OF 1	9381073
			NEXT HIGHER ASSEMBLY: 9381075

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fails to withstand the minimum breaking strength	1/	3.2	4.5.5
102	Length of wire	Level III	3.2	Gage
Minor				
201	Wire ends not fused	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

1/ Ten (10) parts from each lot shall be tested per 4.5.5. If one or more samples fail to meet the requirement, the lot shall be rejected.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.22	Wood, Insert (Prior to Loading XM816 Illuminant Composition)	SHEET 1 OF 1	12953387
			NEXT HIGHER ASSEMBLY 12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level III	1/	Gage
102	Thickness	Level III	1/	Gage
103	Wood grain not perpendicular to insert length	Level III	1/	Visual
104	Wood moisture content incorrect		3.4 1/, 2/	Test
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:
1/ See Note 2 of drawing. 2/ One sample shall be selected for this test prior to the beginning of each day's canister loading.

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4.5 Testing.

4.5.1 Hardness test of illuminant canister. Hardness reading shall be taken on the outside surface, approximately one inch from the open end.

4.5.2 Hardness test of support plate. Hardness reading shall be taken on the outside surface between the spot welds.

4.5.3 Test for adequate weld strength. Test for adequate weld strength shall be conducted by pulling the plate, support from the canister. No more than one undersize weld button is permitted on each sample tested. More than one weld button which measures less than 0.125 inches diameter shall be cause for rejection of the lot.

4.5.4 Surface roughness. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four (4) power magnification may be used in performing visual examination.

4.5.5 Pull test of suspension wire. The test sample shall be assembled on a universal testing machine and the load applied in an axial direction until the required pull force is exerted. The rate of travel of the machine shall be 2 inches per minute maximum.

4.6 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of level C of MIL-STD-1169.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The components covered by this specification are intended for assembly to M853A1 Illuminating and XM816 Infrared, illuminating cartridges for use on the 81mm mortar.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Subject term (key word) listing.

Canister
Cartridge
Illuminant
Strength, weld

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AF75)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70543B (AR)	940816

3. DOCUMENT TITLE: PROJECTILE, 81MM, ILLUMINATING, M853A1 PROJECTILE, 81MM, INFRARED ILLUMINATING XM 816 METAL PARTS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial: (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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